

DTF Transfer Authority (DTFTA) Shopify POD Fulfillment System

1. Introduction

This document explains what we need to build for the DTFTA system in simple terms. It's like a blueprint for the software. The goal is to make it easy for everyone – the client and the developers – to understand what the system does, how it works, and what's required.

1.1 Purpose

We are building a system that helps DTFTA act as a hidden (white-label) partner for Shopify store owners (called merchants or partners). DTFTA will print and ship custom clothing items (like t-shirts and hoodies) on demand when customers order them from these stores.

- **Shopify App:** A tool that merchants install in their Shopify store to connect to DTFTA.
- **DTFTA CRM (Dashboard):** A separate online tool for DTFTA's team to manage orders, print items, ship them, and track everything. It's like a control center.

This system is similar to Printify: Merchants design and sell custom products in their stores, but DTFTA handles the making and shipping without showing their own brand. The app will be public (anyone can install it), charge monthly fees, and bill per order/store.

1.2 Scope

What we will build:

- A Shopify embedded app (it opens inside the Shopify admin panel).
- A separate CRM/dashboard for DTFTA to handle production, shipping, and analytics.
- Connections between Shopify stores, the app, and the CRM using APIs (like digital bridges for data).
- Support for print-on-demand (POD) for specific clothing items.
- White-label shipping (packages look like they come from the merchant, not DTFTA).
- Handling orders from multiple stores, partial fulfillments, cancellations, and errors.

What we won't build (out of scope):

- The merchant's storefront customization tools (merchants handle that in their own Shopify store).
- Inventory storage (since it's POD, items are made only after an order).
- Payment processing for DTFTA's fees (assume it's handled separately, like via Stripe).

1.3 Definitions (Simple Explanations)

- **DTF (Direct-to-Film):** A printing method for custom designs on clothes.
- **POD (Print-on-Demand):** Making products only when ordered, no pre-stocking.
- **White-Label:** DTFTA stays invisible – packages, slips, and labels use the merchant's brand.
- **SKU:** A unique code for each product variant (e.g., DTFTA-APP-TEE-NL-6210-BLK-XL means a black XL unisex tee from Next Level).

- **Fulfillment:** The process of making, packing, and shipping an order.
 - **App:** The software merchants install in Shopify.
 - **CRM/Dashboard:** DTFTA's private tool to see and manage all orders.
 - **Merchant/Partner:** Shopify store owners who use DTFTA for fulfillment.
 - **Job:** An internal task in DTFTA's system for processing one order (like "print this design on a tee").
 - **API:** A way for software to talk to each other (e.g., Shopify sends order info to DTFTA).
 - **Webhook/Callback:** Automatic notifications when something happens (e.g., new order alert).
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2. Overall Description

2.1 How the System Works (High-Level Overview)

Imagine a merchant sells custom t-shirts in their Shopify store. Here's the flow:

1. **Setup:** Merchant installs the app. It creates a "DTFTA location" in their Shopify (set to US for deliveries). They set up their brand info (like return address and support contact).
2. **Product Listing:** The app shows DTFTA's products (e.g., tees, hoodies). Merchants add these to their store with custom designs. All products are customizable (images, text, full options like Printify).
3. **Order Comes In:** Customer buys a custom item. Shopify creates a "fulfillment order" and assigns it to DTFTA.
4. **Notification to DTFTA:** Shopify sends an alert (via webhook/callback) to the app/CRM.
5. **DTFTA Processes:**
 - CRM fetches order details.
 - Validates (checks if SKU is correct, artwork is provided, etc.).
 - If good, creates a "job" in the dashboard.
 - DTFTA team prints the design on the clothing.
 - Generates shipping label and packing slip (with merchant's brand).
 - Ships the item.
6. **Update Shopify:** CRM sends back tracking info to Shopify. Order marked as fulfilled, customer gets notified.
7. **Other Features:**
 - Merchants can partially fulfill (e.g., they ship some items, DTFTA does others).
 - Cancellations: If merchant cancels, DTFTA stops the job.
 - Analytics: Dashboard shows pending/fulfilled orders, stock levels (if any), and store-wise reports.
 - Billing: Monthly subscriptions + per-order fees, invoiced per store.

DTFTA provides the blank clothes and does the printing/shipping. Merchants provide designs/artwork. Inventory is managed by DTFTA (but minimal since POD).

2.2 Users and Roles

- **Merchants (Store Owners):** Install app, set up products, monitor orders in Shopify. They don't access DTFTA's CRM.
- **DTFTA Team (Ops/Admin):** Use CRM/dashboard to view orders, process jobs, ship, and see analytics.
- **Customers:** End-buyers on merchant's store – they see nothing about DTFTA.

2.3 Assumptions

- DTFTA is based in the US and handles only US deliveries.
 - Merchants handle customer support and returns.
 - All products are POD – made after order.
 - SKUs are defined and owned by DTFTA.
 - The app syncs DTFTA's products to merchants' stores, but not vice versa.
 - For validations/errors: If something's missing (e.g., no artwork), flag it and notify merchant.
 - Shipping: DTFTA chooses methods (e.g., USPS, UPS) and generates labels. Needs customer address and method from order.
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3. Functional Requirements (What the System Must Do)

These are the main features, explained step-by-step.

3.1 App Installation and Setup

- When a merchant installs the app (via Shopify App Store):
 - Ask for permission (OAuth) to access their store.
 - Create a "Fulfillment Service" and "Location" in Shopify (named DTFTA, location: US).
 - Store secure access token.
 - Merchant sets up profile: Brand name, return address, support contact (for packing slips).
- App can be installed on multiple stores.
- On uninstall: Clean up everything (remove location, delete data).

3.2 Product Management

- App lists DTFTA's products:
 - Unisex Tee (Next Level 6210)
 - Long Sleeve Tee (Next Level 3601)
 - Heavy Blend Hoodie (Gildan 18500)
 - Heavyweight Hoodie (Cotton Heritage M2650CH)
- Each has variants (colors, sizes) with DTFTA SKUs (e.g., DTFTA-APP-TEE-NL-6210-BLK-XL).
- Merchants can add these to their store.
- All products support full customization: Image upload, text fields, placements (front/back, e.g., 12x16 inches).
- Sync DTFTA products into the app (not merchant's products into app).
- No inventory tracking in app (POD model).

3.3 Order Handling

- **When order placed in merchant's store:**
 - If item has DTFTA SKU and properties (e.g., dtfta_type=APPAREL_POD, garment details, print plan, artwork URLs).
 - Shopify sends notification to app/CRM.
- CRM fetches order via API.
- **Validate:** Check SKU matches, artwork provided, sizes/colors valid, print dimensions okay.
 - If fails (e.g., missing artwork, invalid SKU like wrong size): Set status to "Exception" or "Artwork Needed", notify merchant via email or Shopify note.
- **Create "Job" in CRM:** Shows order details, status (New, In Production, Shipped).
- **Support partial fulfillments:** Merchant can fulfill some items themselves.

3.4 Production and Shipping

- **In CRM Dashboard:**
 - List all pending/fulfilled orders from all stores.
 - Show details: Garment, design, customer address.
 - Team marks job as "In Production" when printing.
 - Generate white-label packing slip (merchant's brand, support info).
 - Generate shipping label (DTFTA handles, using address and method from order).
 - Ship item, add tracking number.
- **Push updates to Shopify:** Mark as fulfilled, add tracking.

3.5 Cancellations and Exceptions

- If merchant cancels in Shopify: Notify CRM, cancel job (or flag if already printing).
- Handle duplicates: Ignore repeat notifications.
- Exceptions: Missing data → Notify merchant to fix.

3.6 Analytics and Reporting

- **In CRM:**
 - View orders by store.
 - Pending, fulfilled, stocks (if tracking blanks).
 - Simple reports: Orders per store, total sales.

3.7 Billing

- Monthly subscription for app.
- Per-order fees, invoiced store-wise.

3.8 Security and Tech

- Secure data (encrypt tokens).
 - Handle webhooks: For uninstalls, order changes.
 - APIs: For fetching orders, creating fulfillments.
 - Edge cases: Partial ships, post-ship tracking updates.
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4. Non-Functional Requirements (How the System Should Perform)

- **Easy to Use:** Simple interfaces, like Printify.
 - **Reliable:** Handle many orders without crashing.
 - **Secure:** Protect data, verify notifications.
 - **Fast:** Respond to alerts quickly (under 5 seconds).
 - **Scalable:** Work for multiple stores/orders.
 - **Mobile-Friendly:** Dashboard works on phones.
 - **Testing:** Must pass tests like install → order → ship → uninstall.
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5. Dependencies

- Shopify account and APIs.
- Hosting for app/CRM (e.g., AWS or Heroku).
- Tools for printing/shipping (assume DTFTA has them).
- Reference: Printify for customization frontend behavior.

This SRS covers everything based on the documents and Q&A. If anything changes, we can update it!
